Document Control

Title			
Pre-Op Assessment Guidelines			
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Version	**Date Issued**	**Status**	**Comment / Changes / Approval**
0.1	Nov 14	Draft	Initial version for consultation
1.0	Jan 15	Final	Approved by Consultant Anaesthetist (Dr Simon Hebard) and Published on Bob
2.0	Apr 2021	Draft	Re-written and re-drafted in new Trust format by Consultant Anaesthetist (Dr Simon Hebard), approved for publication with minor revisions after review by Clinical Reference Group
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Superseded Documents Pre-Operative Assessment Guidelines V1.0 02Feb15 Use of Routine Pre-Operative Tests for Elective Surgery (merged with this document) Pre-operative Assessment ECG/Bloods Referral Guidelines (merged with this document)			
Issue Date Apr 2021	**Review Date** Apr 2024	**Review Cycle** Three years	
Consulted with the following stakeholders: (list all) • **Pre-op Anaesthetic Consultants** • **Pre-Op Assessment Lead Nurse** • **Booked Admissions Manager (Lauren Oke)**			
Approval and Review Process • Presented and discussed at Anaesthetics and Surgery Clinical Governance Meeting (covering all surgical specialties) • Noted at Divisional Governance • Approved Clinical Reference Committee			
Local Archive Reference G:\Anaesthetists **Local Path** Policies & Guidelines **Filename** Pre-Operative Assessments Guidelines V2.0 Apr 2021			
Policy categories for Trust's internal website (Bob) Anaesthetics, Pre-op assessment, Surgery		**Tags for Trust's internal website (Bob)** Anaesthetics, Surgery, Pre-op assessment	

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1. Purpose

1.1. The purpose of this document is to detail the process for healthcare professionals in performing pre-operative assessment for patients undergoing elective surgery here at North Devon District Hospital.

1.2. The guideline applies to all clinical staff involved in the pre-operative assessment of elective surgical patients.

1.3. Implementation of this guideline will ensure that:

- Patients undergoing elective surgery are assessed and optimised in a timely, standardised, efficient and effective manner and in accordance with best practise.
- Process-related delays to the elective surgical pathway are minimised
- Effective, patient-centred perioperative plans are put in place prior to proceeding to elective surgery
- The perioperative outcomes of our patients are optimised

2. Definitions

AF	Atrial Fibrillation	**MI**	Myocardial Infarction
ASA	American Society of Anaesthesiologists Physical Status Classification System (see Appendix 8.2)	**Nebs**	Nebulisers
BMI	Body Mass Index	**NICE**	National Institute for health and Clinical Excellence
BMS	Bare metal stent	**NIV**	Non-Invasive Ventilation
BP	Blood Pressure	**NSAIDs**	Non-Steroidal Anti- Inflammatory Drugs
BiPAP	Bi-level Positive Airway Pressure ventilation	**NICE**	National Institute for Health and Care Excellence
CABG	Coronary Artery Bypass Grafting	**NIDDM**	Non-Insulin Dependent Diabetes Mellitus
CCF	Congestive Cardiac Failure	**O2**	Oxygen
CPAP	Continuous Positive Airway Pressure	**OSA**	Obstructive Sleep Apnoea
DES	Drug Eluting Stent (DES)	**PE**	Pulmonary Embolism

DM	Diabetes Mellitus	**PFT's**	Pulmonary Function Tests
DVT	Deep Vein Thrombosis	**Plts**	Platelets
ECG	Electrocardiograph	**PND**	Paroxysmal Nocturnal Dyspnoea
ECHO	Echocardiogram	**PPM**	Permanent Pacemaker
FBC	Full Blood Count	**SOB**	Shortness of Breath
GP	General Practitioner	**TCI**	To Come In
HDU	High Dependency Unit	**TFT's**	Thyroid Function Tests
ICD	Implanted Cardiovertor Defibrillator	**TIA**	Transient Ischaemic Attack
IDDM	Insulin-Dependent Diabetes Mellitus	**TURP**	Trans-Urethral Resection of Prostate
INR	International Normalised Ratio	**SVT**	Supraventricular Tachycardia
JW	Jehovahs Witness	**U&E's**	Urea and Electrolytes
LFT's	Liver Function Tests	**Virtual assessment**	Review of clinical notes alone (i.e. no patient present)

3. Responsibilities

Role of the Pre-Op Assessment Nurse

3.1. The Pre-Op Assessment Nurse is responsible for:

- Carrying out a comprehensive assessment of each patient referred to the Pre-Op Assessment Service
- Where this assessment is carried out through either telephone or video consultation, ensuring that appropriate Trust procedure is followed ("Telephone Appointment Standard Operating Procedure" and "Video Consultation Standard Operating Procedure")

- Acting on their clinical findings in accordance with this guideline so as to ensure that each patient is investigated and optimised prior to surgery
- Collating and communicating to the pre-op anaesthetist and/or listing surgeon the outcome of specialist investigations and/or specialist opinions, along with any additional documentation (e.g. copies of records from other Trusts) as necessary in order for the clinicians to confirm each patient's perioperative plan
- Documenting and communicating each patient's perioperative plan to all involved

Role of the Pre-Op Anaesthetist

3.2. The pre-op anaesthetist is responsible for:

- Providing advice on the perioperative management of elective surgical patients via review of clinical notes alone ("virtual assessment").
- Assessing high-risk elective surgical patients face-to-face where review of clinical notes alone is deemed insufficient to achieve the above ("face to face assessment").
- Making referrals for further investigation and/or specialist opinion as required in order to formulate each patient's perioperative plan

Role of the Listing Surgeon

3.3. The surgeon is responsible for:

- Providing advice on the surgical management of elective patients pre-operatively.

Role of Clinical Effectiveness Group (CEG)

3.4. The Clinical Effectiveness Group is responsible for:

- Approving this clinical guideline

Role of Medical Director

3.5. The Medical Director has executive responsibility for

- Ensuring compliance with this guideline

4. The Timing of Pre-op Assessment

4.1. In January 2015, in line with previous best practise guidance from the NHS Modernisation Agency, the Trust moved to a model of early pre-operative assessment as standard (i.e. as soon as possible after the decision to treat).

4.2. This change was intended to facilitate the efficiency and safety of the elective surgical pathway, particularly in the context of short-notice theatre bookings.

4.3. In order to support this change, the following safety nets were put in place:

- When a patient's pre-op assessment is complete, the **pre-op nurse** will:

  (i) Ensure that a perioperative plan is documented in the patient's pre-op assessment record (paper or electronic)

- When the **booked admissions team** issue a TCI date (new booking or change of existing booking), they will do the following:

  (ii) Confirm whether the patient has had their pre-op assessment (if not then book a pre-op appointment). Pre-op assessment and associated investigations are currently valid for six months.

  (iii) Confirm that the patient has a perioperative plan documented in their pre-op record (if not then contact pre-op assessment)

  (iv) Contact the patient and confirm that there has been no change in their health or social circumstances by asking if there has been:

  1. **Admission to hospital**
  2. **Significant illness (hospital consultation or a new treatment)**
  3. **Change in medication**
  4. **Relevant change in social circumstances**

  If the answer is "Yes" to any of the above questions then contact pre-op assessment, otherwise proceed as below:

  (v) Contact any teams that need to be informed of the patient's TCI date and make allowance for any specific logistical requirements as documented in the patient's perioperative plan (e.g. stopping antiplatelet or anticoagulant medication prior to surgery).

5. The Pre-op Assessment Process

5.1. All patients should be assessed utilising current Trust pre-operative assessment documentation (paper or electronic).

5.2. Local guidance with regard to routine pre-operative testing (e.g. bloods, ECG, etc.) can be found in the Appendix to this document and is based on current NICE guidance.

5.3. The findings of each patient's nurse pre-operative assessment should be utilised to guide any necessary further investigation and management including the need for referral to the anaesthetic high-risk clinic for either virtual or face to face assessment. See the Appendix of this document for current guidance on who or what to refer.

5.4. Pre-op anaesthetists within the anaesthetic high risk clinic service will utilise the nurse pre-op findings to direct their virtual or face to face patient assessment in order to:

- Assess and optimise the patient's physical status and underlying health conditions including the arrangement of further investigation, treatment and onward specialist referral as clinically indicated
- Assess and discuss the patient's perioperative risk (see the Appendix of this document for current spectrum of risk assessment tools) in order to inform a shared decision with the patient and the listing surgeon regarding the proposed procedure
- Formulate an appropriate perioperative plan

5.5. Where the magnitude of perioperative risk is such that the pre-op anaesthetist feels that the intended procedure is not in the best interests of the patient, they will:

- Communicate this assessment to the listing surgeon

5.6. Should the listing surgeon still wish to proceed then they may:

- Request an independent assessment of perioperative risk from a second pre-op anaesthetist here at NDDH
- Utilise this independent risk assessment to inform a final shared decision between the original pre-op anaesthetist, the patient and themselves (e.g. at a face to face meeting) as to whether to proceed to surgery or not

5.7. On the rare occasion that disagreement still persists between the pre-op anaesthetist, the listing surgeon and the patient despite the above steps, the listing surgeon should:

- Offer referral to an alternative NHS Trust for a second opinion

5.8. Once each patient's pre-operative assessment and any necessary optimisation are complete, the pre-op assessment nurse will document and communicate their final perioperative plan. This plan should consider the following:

- Medications management (e.g. ACE-inhibitors, angiotensin receptor blockers, antiplatelets, anticoagulants, diabetes medications, immunosuppressants)
- Seniority of anaesthetist +/- need for sub-specialist skills (e.g. awake surgery under regional block, awake fibreoptic intubation)
- Seniority of surgeon
- Need for special equipment
- Need for a HDU or high acuity bed
- Significant patient alerts (e.g. difficult airway, obesity, diabetes, previous anaphylaxis, etc.)
- Restrictions to theatre environment (e.g. main theatres only)
- Any "Reasonable Adjustments" pathway plan
- Suitability for evening or week-end list (see Appendix of this document for current acceptance criteria)
- Appropriate surgical pathway:

➢ Daycase +/- overnight stay
➢ Inpatient

The following information should be taken into account when determining the appropriate surgical pathway:

Patient demographic, contact and social information

5.9. Each patient's demographic, contact and social information should be reviewed and recorded routinely for all patients.

5.10. Both the level of patient supervision and the communications infrastructure available post-discharge must be consistent with the pathway documented (e.g. compliant with Trust day surgery criteria)

Intended procedure

5.11. The intended surgical procedure should be reviewed and recorded routinely for all patients. The information source for this data should be the patient's surgical listing form or electronic equivalent unless agreed otherwise.

5.12. Classification by NICE Surgery Grade (see Appendix for guide to NICE Surgery Grades) can be used to assess whether each patient is on the appropriate surgical pathway as below:

- **Minor or Intermediate**
  - Accepted for day surgery
- **Major or Complex**
  - Accepted for day surgery only if approved by the listing surgeon and a pre-op anaesthetist

Physical Status (ASA grade)

5.13. Pre-operative assessment findings should be utilised to assign and document an ASA grade for each patient (see Appendix for guide to ASA classification).

5.14. Classification by ASA grade can be used to assess whether each patient is on the appropriate surgical pathway as below:

- **ASA 1 & 2**
  - Accepted for day surgery
- **"Stable ASA 3"** (i.e. chronic disease states with stable symptoms and functional capacity)
  - Refer pre-op anaesthetist
    - May be restricted to main theatres operating list only
    - May require overnight stay or inpatient surgery
- **"Unstable ASA 3"** (i.e. fluctuating health/functional status)
  - Refer pre-op anaesthetist as unlikely to be suitable for day surgery
- **ASA ≥ 4**
  - Refer pre-op anaesthetist as highly unlikely to be suitable for day surgery

6. Monitoring Compliance with and the Effectiveness of the Guideline

Standards/ Key Performance Indicators

6.1. Key performance indicators comprise:

- Cancellation on the day of surgery
- Clinical incidents and near misses

Process for Implementation and Monitoring Compliance and Effectiveness

6.2. Final approved guideline to be cascaded to all pre-op assessment nursing staff and pre-op anaesthetic Consultants and published on the Trust Policies site on BOB. Link to guideline document to be created from Pre-op intranet page on BOB.

6.3. Detail here the monitoring process:

- Day of surgery cancellations are routinely monitored and any recurring themes highlighted and addressed via Theatre Users Group (TUG)
- Clinical incidents are routinely reported and investigated via the Trust's DATIX reporting system
- Any necessary improvements/revisions to this guideline will be the responsibility of the Consultant and Lead Nurse for pre-op assessment here at NDDH.

7. Associated Documentation

- Perioperative Diabetes guidelines
- Perioperative Anaemia guidelines
- Perioperative Anticoagulation Guidelines
- Perioperative Management of Rheumatology Medications Guideline
- Pregnancy Screening Prior to Surgery – Standard Operating Procedure
- Paediatric Reasonable Adjustments Pathway for Day Surgery Standard Operating Procedure
- Telephone Appointment Standard Operating Procedure
- Video Consultation Standard Operating Procedure
- Agreed Surgical Blood Ordering Schedule

8. Appendix

Routine pre-operative tests for elective surgery

8.1. Purpose

The purpose of this guidance is to help practitioners to determine the minimum level of testing that a patient should have prior to undergoing elective surgery at North Devon District Hospital.

8.2. Background

This guidance covers routine preoperative tests for people aged over 16 who are having elective surgery.

It is a local adaptation of the 2016 NICE Guideline (NG45 “Routine preoperative tests for elective surgery”) available at http://www.nice.org.uk).

Many apparently healthy people are tested before surgery to check for undetected conditions that might affect their treatment. This can provide a benefit where test results yield additional information that cannot be obtained from a patient history and physical examination alone. However, excessive preoperative testing can cause significant anxiety, delays in treatment and unnecessary, costly and possibly harmful treatments when false positive results are obtained. Even genuinely abnormal results often do not result in any significant change in perioperative management in relatively healthy people.

This guideline recommends when ‘routine’ preoperative tests should be performed. The word ‘routine’ implies the generic testing of people in whom there is no indication for the test, other than that they may have an operation. Here at NDDH, this is meant as a minimum standard and does not replace the autonomy and clinical acumen of the practitioner (i.e. if your clinical assessment suggests further testing, please do so).

8.3. Consider existing medicines

Always take into account the medicines patients are taking when deciding on preoperative testing. Common examples would include:

- Blood pressure medications and diuretics: Check U&E
- Antiplatelet medications: Check FBC
- Anticoagulants: Check FBC/coagulation
- Diabetes medications: Check HbA1c
- Thyroid medications: Check TFT's
- Treatment for anaemia: Check FBC

8.4. Basic tests

Test	ASA 1	ASA 2	ASA 3 or ASA 4
Minor surgery (**examples**: excising skin lesion; draining breast abscess)			
Full blood count	Not routinely	Not routinely	Not routinely
Haemostasis	Not routinely	Not routinely	Not routinely
Kidney function	Not routinely	Not routinely	Consider in people at risk of AKI[1]
ECG	Not routinely	Not routinely	Consider if no ECG results available from past 12 months
Lung function/arterial blood gas	Not routinely	Not routinely	Not routinely
Intermediate surgery (**examples**: primary repair of inguinal hernia; excising varicose veins in the leg; tonsillectomy or adenotonsillectomy; knee arthroscopy)			
Full blood count	Not routinely	Not routinely	Consider for people with cardiovascular or renal disease if any symptoms not recently investigated
Haemostasis	Not routinely	Not routinely	Consider in people with chronic liver disease • If people taking anticoagulants need modification of their treatment regimen, make an individualised plan in line with local guidance • If clotting status needs to be tested before surgery (depending on local guidance) use point-of-care testing[2]
Kidney function	Not routinely	Consider in people at risk of AKI[1]	Yes
ECG	Not routinely	Consider for people with cardiovascular, renal or diabetes comorbidities	Yes
Lung function/arterial blood gas	Not routinely	Not routinely	Consider seeking advice from a senior anaesthetist as soon as possible after assessment for people who are ASA grade 3 or 4 due to known or suspected respiratory disease
Major or complex surgery (**examples**: total abdominal hysterectomy; endoscopic resection of prostate; lumbar discectomy; thyroidectomy; total joint replacement; lung operations; colonic resection; radical neck dissection)			
Full blood count	Yes	Yes	Yes
Haemostasis	Not routinely	Not routinely	Consider in people with chronic liver disease • If people taking anticoagulants need modification of their treatment regimen, make an individualised plan in line with local guidance • If clotting status needs to be tested before surgery (depending on local guidance) use point of care testing[2]
Kidney function	Consider in people at risk of AKI[1]	Yes	Yes
ECG	Consider for people aged over 65 if no ECG results available from past 12 months	Yes	Yes
Lung function/ arterial blood gas	Not routinely	Not routinely	Consider seeking advice from a senior anaesthetist as soon as possible after assessment for people who are ASA grade 3 or 4 due to known or suspected respiratory disease

AKI, acute kidney injury
[1]See recommendation 1.1.8 of the NICE guideline on acute kidney injury
[2]Note that currently the effects of direct oral anticoagulants (DOACs) cannot be measured by routine testing.

8.5. Urine tests

Consider microscopy and culture of midstream urine sample before surgery if the presence of a urinary tract infection would influence the decision to operate.

This would include orthopaedics (using metalwork) and urology surgery.

8.6. HbA1c testing

We have elected to continue screening selected patients for diabetes due to the frequency of abnormal results encountered within our unit.

See Trust Perioperative Diabetes Guidelines for current screening criteria.

8.7. Pregnancy tests

See Trust standard operating procedure: "Pregnancy Screening Prior to Surgery – Standard Operating Procedure"

8.8. Blood Ordering

Any necessary "Group and Save" blood samples should be sent in accordance with the Trust's "Agreed Surgical Blood Ordering Schedule".

Anaesthetic referral guidelines

8.9. The following tables provide guidance for referral to the pre-op anaesthetists in the Anaesthetic High Risk Clinic for either virtual (review of clinical notes alone) or face to face assessment:

Body Mass Index (BMI)	
Aim: to identify those at elevated perioperative risk due to abnormal BMI and associated conditions.	
Obesity	
Obesity results in technical challenges with regard to the safe conduct of both anaesthesia and surgery, often necessitating the rostering of senior staff, the availability of specialist equipment and the provision of extra theatre time. As the degree of obesity increases (BMI ≥ 40) or where it is complicated by the metabolic syndrome (obesity with hypertension and/or diabetes), patients are at increased risk of perioperative complications and it is not uncommon to request the availability of a postoperative HDU bed.	
Criteria	Action Plan
BMI ≥ 40	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic** Notify theatre team of weight/BMI via alert on the Trust's electronic theatre management system and request bariatric equipment
BMI ≥ 50	**Refer for face to face assessment in anaesthetic high-risk clinic** Notify theatre team of weight/BMI via alert on the Trust's electronic theatre management system and request bariatric equipment
Low Body Weight	
Low body weight is associated with increased risk of perioperative complications.	
Criteria	Action Plan
BMI < 18.5	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic** Notify theatre team of weight/BMI via alert on the Trust's electronic theatre management system

Cardiac History

Aim: to identify any uncontrolled cardiovascular disease.	
Ischaemic Heart Disease	
Cardiac pain is identified as a pressure on the chest or crushing central chest pain. It may radiate to the left arm, neck or jaw. It is usually brought on by cold weather, wind, exercise, heavy meal. It is usually relieved by rest or GTN.	
Criteria	Action Plan
Myocardial infarction within last 6 months or angina causing limitation of normal daily activity or incapacitation.	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Poor symptom control or progression in symptoms	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Symptoms controlled, only on strenuous exertion	ECG
Angiogram previously	Chase results
Previous angioplasty/stents	Check medication list for antiplatelet therapy (aspirin/clopidogrel/prasugrel/ticagrelor) - management as per Trust Perioperative Anticoagulation and Antiplatelet Guideline.

High Blood Pressure	
Criteria	Action Plan

BP ≥ 180/110	Repeat measurement twice and accept lowest value.
	If BP remains ≥ 180/110 by lowest value:
	• Re-check in community (home or via GP practice)
	If BP remains ≥ 180/110 in community:
	• Elective surgery:
	  ○ Defer (six weeks) - Refer to GP
	• Urgent surgery:
	  ○ **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

BP < 180/110	Proceed

Heart Murmur (especially if history of previous Rheumatic Fever)	
Criteria	Action Plan
New murmur/ asymptomatic	ECHO. **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

New murmur/ symptomatic	ECHO. **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May need to defer and refer to cardiology
Old murmur/ no change in symptoms	Chase ECHO (must be within last 12 months for moderate/severe stenotic lesions) and cardiology letters.
Old murmur/ progression of symptoms	ECHO. **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May need to defer and refer to cardiology
Associated arrhythmias e.g. Atrial fibrillation	Check medications, especially anticoagulants - management as per Trust Perioperative Anticoagulation and Antiplatelet Guideline.

Arrhythmias	
• **Atrial Fibrillation**	
Criteria	Action Plan
Rate controlled	Check medications, especially anticoagulants - management as per Trust Perioperative Anticoagulation and Antiplatelet Guideline.
New AF <u>or</u> Rate/symptoms uncontrolled	**Refer to arrhythmia care coordinator (see Appendix of this document for referral pathway) <u>and also</u>:** **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
• **SVT**	
Criteria	Action Plan
Rate controlled	Nil
• **Ventricular Arrhythmias**	
Criteria	Action Plan
Rhythm & symptoms controlled	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Rhythm & symptoms <u>not controlled</u>	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May need to defer and refer to cardiology
• **Bradycardia and/or heart block**	
Criteria	Action Plan

Asymptomatic with: • Sinus bradycardia in athletes or those on beta blockers • First degree heart block • Bi or tri fascicular block	Proceed
Symptomatic (faints/blackouts/shortness of breath)	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May need to defer and refer to cardiology
Second degree heart block	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May need to defer and refer to cardiology
Complete heart block	Admit under cardiology
• **Permanent Pacemaker (PPM) in situ**	
Criteria	Action Plan
PPM in-situ	ECG, check indication and date of last check (must be within last 12 months)
Symptoms not controlled	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May need to defer and refer to cardiology
• **Implanted Cardiovertor Defibrillator (ICD) in situ**	
Criteria	Action Plan
ICD in-situ	ECG, check indication and date of last check (must be within last 12 months). Notify cardiorespiratory department of TCI date (to arrange for perioperative device deactivation/ reactivation on day of surgery) **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Symptoms not controlled	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May need to defer and refer to cardiology

Cerebrovascular disease	
Criteria	Action Plan
Stroke/TIA within last three months	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** Will need to be deferred unless clinically urgent.
Stroke/TIA more than three months ago	Chase carotid doppler if done Chase ECHO if done Review medications (e.g. antiplatelets and/or anticoagulation) - management as per Trust Perioperative Anticoagulation and Antiplatelet

	Guideline.

Shortness of breath (Cardiac)	
It can be difficult to distinguish between SOB due to cardiac or respiratory disease and in some cases patients have both. However, it is an important classification which results in quite different treatment.	
Criteria	Action Plan
SOB on exertion (>two flights) and no progression of symptoms	Chase ECHO if done
SOB on minimal exertion (<one flight) and no progression of symptoms	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic** Chase ECHO (request if not done)
SOB on minimal exertion and worsening symptoms, orthopnea or PND	Need medical optimisation: ECHO **Refer for face to face assessment in anaesthetic high-risk clinic.** May need to defer and refer to cardiology

Respiratory Disease	
COPD/ Bronchitis/ Asthma	
Criteria	Action Plan
Acute exacerbation or infection	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** Will need to defer for four weeks unless clinically urgent.
Mild symptoms	Proceed
SOB on exertion (>two flights) and no progression of symptoms	Chase pulmonary function tests (PFT's) if done
SOB on minimal exertion (less than one flight) and no progression of symptoms	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** Chase PFT's (request if not done)

Severe disease/home nebs/home oxygen/NIV or SOB on minimal exertion and worsening symptoms, orthopnea or PND	**Refer for face to face assessment in anaesthetic high-risk clinic.** Can this patient be improved? Is surgery appropriate? May need to defer and refer to chest physician.

Tuberculosis	
Criteria	Action Plan
Treatment in last six months/ productive cough/ active TB	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May not be suitable for surgery

Obstructive Sleep Apnoea (OSA)	
Criteria	Action Plan
Treated (i.e. CPAP/ BiPAP)	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** Patient to bring in own CPAP
Untreated	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May need HDU post-op.

Renal Disease	
Criteria	Action Plan
Mild renal dysfunction	Proceed
Chronic kidney disease (CKD) stage 4 (eGFR 15-29 ml/min)	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.**
Chronic kidney disease (CKD) stage 5 (eGFR <15 ml/min)	**Refer for face to face assessment in anaesthetic high-risk clinic.** May need discussion with nephrology
Haemo or peritoneal dialysis	**Refer for face to face assessment in anaesthetic high-risk clinic.** Likely to need discussion with nephrology May not be suitable for surgery here at NDDH

Endocrine Disease	
Diabetes Mellitus	
Criteria	Action Plan

All patients	Management as per Trust perioperative diabetes guidelines
Poor diabetic control (HbA1c > 69 mmol/mol)	**Refer Diabetes Specialist Nurse team** (see Appendix of this document for referral pathway and form) **and also:** **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Thyroid Disease	
Criteria	Action Plan
Symptoms controlled with recent blood test	Chase TFT's
Poor symptom control or no recent blood tests	Check TFTs, may need treatment optimisation
Complication of thyroid surgery e.g. laryngomalacia, vocal cord palsy.	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Gastro-oesophageal reflux disease (GORD)	
Criteria	Action Plan
Known hiatus hernia or GORD controlled on medical treatment	Document. Patient to continue treatment including day of surgery.
Regular GORD symptoms or reflux at night/ lying flat or bending forward	Document. Anaesthetist to consider antacid pre-med on the day of surgery.

Neurological & Neuromuscular Disease	
Epilepsy	
Criteria	Action Plan
Well-controlled epilepsy	Take all medications pre-operatively.
Poorly-controlled epilepsy	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Uncontrolled seizures with no treatment	**Refer for face to face assessment in anaesthetic high-risk clinic** Likely unfit for surgery and will need to defer and refer to neurology

Multiple Sclerosis (MS)	
Criteria	Action Plan
Well with minimal limitation of physical activity	Proceed
Moderate or severe limitation of activity	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Myasthenia Gravis or Muscular Dystrophy	
Criteria	Action Plan
All patients	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Inherited/genetic disorders & congenital syndromes	
Down's Syndrome	
Criteria	Action Plan
All patients with Down's syndrome	**Notify theatre team via alert on the Trust's electronic theatre management system**
Down's with neck problems	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Down's with cardio-respiratory problems	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Others	
All patients	Document details of condition **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Haematological problems	
Anaemia	
Criteria	Action Plan
Patient for major surgery (e.g. major colorectal surgery, hysterectomy or major joint surgery)	Management as per Trust perioperative anaemia guidelines
Other surgery	Request haematinics and notify GP of results **If severe anaemia (Hb < 100g/L):** • **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic** • May need deferral for optimisation or investigation.

All African, Afro-Caribbean, Asian, East-Mediterranean & Middle-Eastern patients	• Test for Sickle cell status if not previously documented • Sickle Trait: Document • Sickle Disease: **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Coagulopathy	
Criteria	Action Plan
INR > 1.4 or Plts < 100	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.** May need discussion with haematology +/- deferral for investigation/optimisation
Known bleeding disorder	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Pro-coagulant	
Criteria	Action Plan
Patient taking regular anticoagulant medication	Management as per Trust Perioperative Anticoagulation and Antiplatelet Guideline

Transfusion	
Criteria	Action Plan
Jehovah's Witness (or others who do not accept transfusion of blood products)	Ensure up to date advanced directive filed in patient's medical notes **Refer for face to face assessment in anaesthetic high-risk clinic**

Liver Disease	
Hepatitis	
Criteria	Action Plan
Known current or previous Hepatitis B or C	Check FBC, coagulation screen, LFT's, U&E's. Place alert on electronic theatre management system

Chronic Liver Disease	
Criteria	Action Plan
All patients	Check FBC, coagulation screen, LFT's, U&E's **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Jaundice	
Criteria	Action Plan
All jaundiced patients	Check FBC, coagulation screen, LFT's, U&E's **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Arthritis	
Rheumatoid arthritis	
Criteria	Action Plan
Simple articular rheumatoid arthritis	Nil
Complex rheumatoid arthritis and/or limitation of mouth opening or cervical spine disease	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Taking immunosuppressants	Management as per Trust guideline - "Perioperative Management of Rheumatology Medications in North Devon"

Arthritis - other	
Criteria	Action Plan
Limitation of mouth opening or neck movement.	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Marked scoliosis affecting respiration.	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Taking immunosuppressants	Management as per Trust guideline - "Perioperative Management of Rheumatology Medications in North Devon"

Lifestyle	
Smoking	
All patients	Document and provide health promotion advice
Alcohol (excessive intake)	
All patients	Document and provide health promotion advice Do FBC, U&E's, LFT's, clotting screen, ECG.
Could patient be pregnant?	
Ask patient if could be pregnant. Document: • Date of last menstrual period, if applicable. • Form of contraceptive currently being used.	Follow Trust Standard Operating Procedure

Anaesthetic History	
Previous anaesthetics	
Criteria	Action Plan
All patients	Check details and document

Problems with previous anaesthetics	
Criteria	Action Plan
Suxamethonium apnoea Malignant hyperpyrexia Significant morbidity during or after anaesthesia (e.g. unplanned admission to inpatient ward or critical care unit) Known difficult intubation	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Post-operative nausea and vomiting (PONV) Take a long time to wake up post-operatively Low blood pressure not requiring post-op anaesthetic/critical care intervention	Check details and document

Family History of Anaesthetic problems	
Criteria	Action Plan
Suxamethonium apnoea Malignant hyperpyrexia Mortality or significant morbidity in a family member as a result of an “allergy” to general anaesthesia	**Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**

Drug History	
Most medications can and should be taken as normal on the day of surgery. Below are a few exceptions. This is not an exhaustive list and if in doubt discuss the case with a pre-op anaesthetist.	
Anticoagulants	Management as per Trust Perioperative Anticoagulation and Antiplatelet Guideline. Likely to need medication stopping perioperatively and, depending on the reason for anticoagulation, may require “bridging therapy” with an alternative anticoagulant.
Antiplatelets	Management as per Trust Perioperative Anticoagulation and Antiplatelet Guideline. Cardiology advice may be required before stopping

	treatment if coronary stent in-situ.
ACE inhibitors & Angiotensin receptor blockers	Stop one day prior to surgery (i.e. instruct patient not to take on the day before or on the day of surgery)
Monoamine oxidase inhibitors (MAOI's)	Older antidepressants. Now rarely used. **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic.**
Diabetic medication	Management as per Trust perioperative diabetes guidelines

ECG: Automated, self analyzing ECG's are the norm now and can flag up abnormalities that are not clinically significant. If in doubt, please refer.	
Need to refer	**Don't need to refer**
Newly diagnosed Atrial fibrillation or flutter Atrial fibrillation or flutter if rate HR > 100	Sinus Rhythm Known Atrial fibrillation or flutter HR<100
Sinus tachycardia >110 Sinus bradycardia <40	Sinus arrhythmia Isolated bradycardia 40-60
Second degree heart block Third degree heart block Bifascicular block (only if symptomatic) Trifascicular block (only if symptomatic) Left bundle branch block	First degree heart block Isolated left or right axis deviation Isolated right bundle branch block
Evidence of definitive infarct or ischaemia Significant ST changes	Non specific P/ QRS/ ST abnormalities
Left ventricular hypertrophy with strain	Isolated left or right ventricular hypertrophy
Multiple ventricular ectopics	Single ectopics (atrial or ventricular) Multiple atrial/supraventricular ectopics

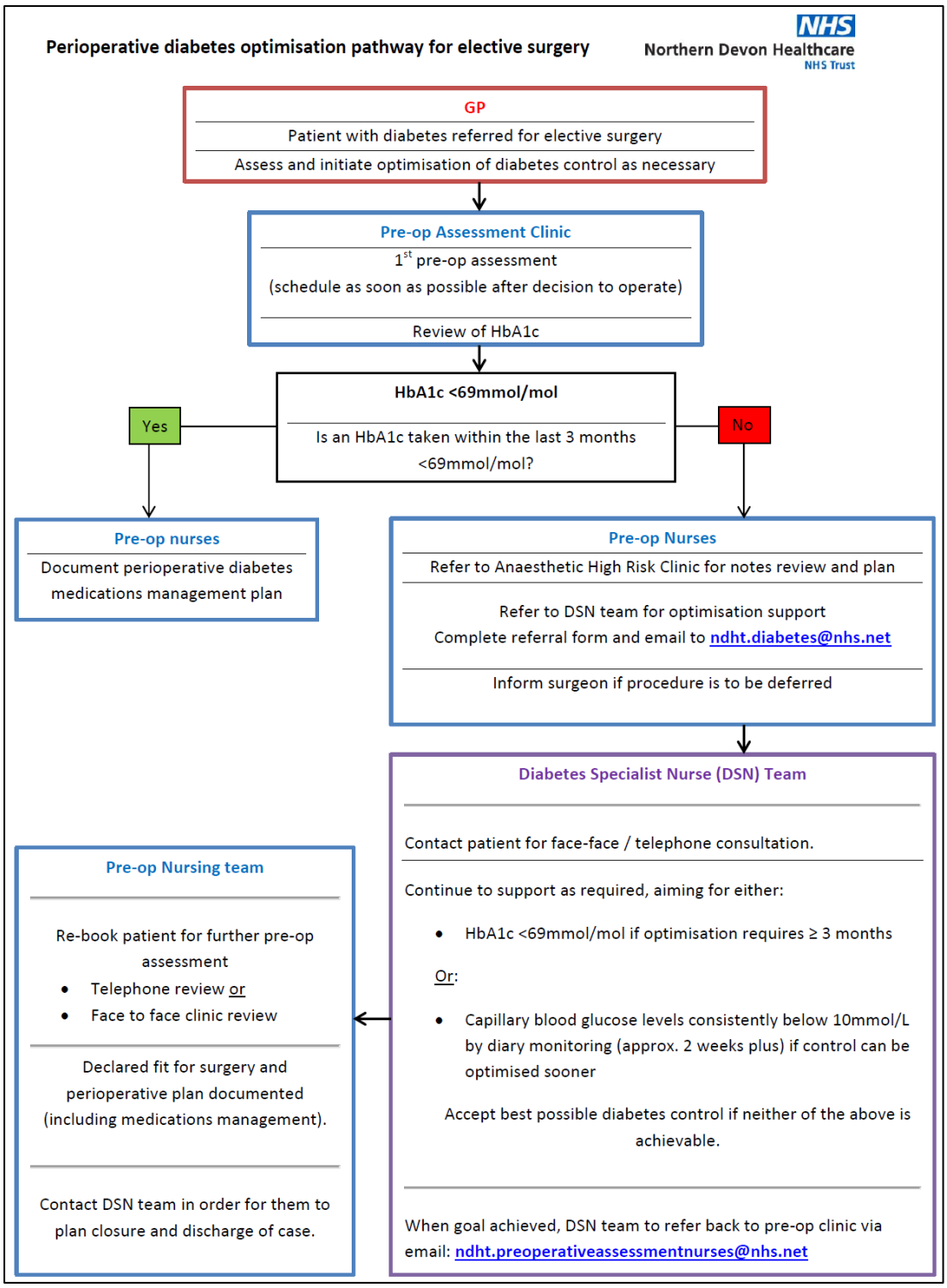
Blood tests: Please refer external to parameters defined below		
Test	**Parameter**	**Note**
Hb	Male<130 female<120	Management as per Trust preoperative anaemia guidelines
HbA1c	>69 (8.5%)	**Refer Diabetes Specialist Nurse team** (see Appendix of this document for referral pathway and form) **<u>and also</u>:** **Refer for virtual assessment (i.e. review of clinical notes) in anaesthetic high-risk clinic**
Platelets	<120 or >500	
WCC	>14	Unless there is evidence of infection

INR APTR	>1.4 >1.4	Unless on warfarin
Sodium Potassium Urea Creatinine	<130 or >148 <3.5 or >5.5 >12 >140	Unless established and stable CKD 3
Bilirubin ALT	>25 >70	

Optimisation pathways

Diabetes

8.10. The following flowchart and referral form should be used to facilitate the optimisation of patients with poor diabetic control prior to elective surgery:


Perioperative diabetes optimisation pathway for elective surgery
NHS
Northern Devon Healthcare
NHS Trust
GP
Patient with diabetes referred for elective surgery
Assess and initiate optimisation of diabetes control as necessary
Pre-op Assessment Clinic
1st pre-op assessment
(schedule as soon as possible after decision to operate)
Review of HbA1c
HbA1c <69mmol/mol
Is an HbA1c taken within the last 3 months <69mmol/mol?
Yes
No
Pre-op nurses
Document perioperative diabetes medications management plan
Pre-op Nurses
Refer to Anaesthetic High Risk Clinic for notes review and plan
Refer to DSN team for optimisation support
Complete referral form and email to ndht.diabetes@nhs.net
Inform surgeon if procedure is to be deferred
Diabetes Specialist Nurse (DSN) Team
Contact patient for face-face / telephone consultation.
Continue to support as required, aiming for either:
• HbA1c <69mmol/mol if optimisation requires ≥ 3 months
Or:
• Capillary blood glucose levels consistently below 10mmol/L by diary monitoring (approx. 2 weeks plus) if control can be optimised sooner
Accept best possible diabetes control if neither of the above is achievable.
When goal achieved, DSN team to refer back to pre-op clinic via email: ndht.preoperativeassessmentnurses@nhs.net
Pre-op Nursing team
Re-book patient for further pre-op assessment
• Telephone review or
• Face to face clinic review
Declared fit for surgery and perioperative plan documented (including medications management).
Contact DSN team in order for them to plan closure and discharge of case.

Perioperative diabetes optimisation pathway for elective surgery

NHS
Northern Devon Healthcare
NHS Trust

Pre-op surgical Referral Form: Patients requiring intense Diabetes Specialist Nurse support

Surname:	**Forename:**	**Title:**
NHS number:	**DOB:**	**Age:**
Address:		
Contact Telephone Numbers: **Home:**	**Mobile:**	**Work:**
Type of diabetes:	**HbA1c:**	**Procedure:**
Medication:		
Date of surgery (if known)		
Other Relevant Information		
Name of Referrer:		**Date:**

Please email form to **ndht.diabetes@nhs.net**

Any queries please contact the team on 01271 322726

Atrial Fibrillation

8.1. The following pathway should be used to facilitate the optimisation of patients with new or poorly controlled (rate > 100) AF prior to elective surgery:

- When patients are diagnosed with new or poorly controlled atrial fibrillation / flutter on ECG during pre-op assessment the pre-op team will contact the Arrhythmia service.
- If immediately available, the Arrhythmia Care Coordinator (ACC) will review the patient in the pre-op assessment clinic.
- If not available immediately, they will plan with the patient a suitable time for an early OP review.
- During OP review symptoms will be assessed and medications reviewed.
- Their CHA2DS2-VASc stroke risk will be assessed to consider if oral anticoagulation is advisable.
- An echocardiogram will be arranged and up to date blood tests, including thyroid function.
- A rate or rhythm control treatment plan will be agreed with the patient.
- If a rate control strategy is agreed and further rate control is required, medication advice is given following local AF treatment guidelines located on 'BOB' and a 24 hour heart monitor arranged to ensure appropriate heart rate control, if required.
- If a rhythm control strategy is agreed arrangements will be made for the patient to be added to the elective DC cardioversion waiting list.
- If further advice is required the ACC with liaise with the 'Cardiologist of the Week' (COW).
- The outcome of the review and treatment plan will be copied to the listing surgeon and pre-op nursing team.

Acceptance Criteria for Week-end or Evening Operating Lists

8.2. Background

The following criteria have been drawn up by the perioperative team (Pre-op Consultant Anaesthetists/Pre-op Nurses/Consultant Orthopaedic Surgeons) in order to facilitate a safe, consistent and efficient approach to the scheduling of patients having elective surgery out of hours (i.e. on week-end and evening operating lists):

- They are not intended to supersede clinical judgement.

- Risk-benefit discussions between surgeons and pre-op anaesthetists are strongly encouraged.

8.3. Acceptance Criteria

ASA grade

- *ASA 1 or 2 - Accepted*
- *"Stable ASA 3" – Accepted by pre-op anaesthetist approval only*

Body Mass Index (BMI)

- *BMI ≤ 35 – Accepted*

Difficult Airway

- *Patients who are documented as having a difficult airway (e.g. requiring fibreoptic intubation) are not accepted*

Critical Care (HDU/ICU)

- *Patients documented as requiring critical care post-operatively are not accepted*

Dementia/Learning Difficulties/Behavioural Issues

- *Patients with significant dementia/learning difficulties/behavioural issues are not accepted*

Perioperative Risk Prediction

8.4. The following tools should be used to facilitate the prediction of perioperative risk and hence inform the process of shared decision making prior to elective surgery:

Perioperative Risk Prediction Tools
ASA (American Society of Anaesthesiologists) Classification System
The ASA Physical Status Classification System has been in use for over 60 years. The purpose of the system is to assess and communicate a patient's pre-anaesthesia medical co-morbidities. The classification system alone does not predict the perioperative risk but, used with other factors (eg, type of surgery, frailty, level of deconditioning), it can be helpful in predicting perioperative risks. Assigning a Physical Status classification level is a clinical decision based on multiple factors. While the Physical Status classification may initially be determined at various times during the preoperative assessment of the patient, the final assignment of Physical Status classification is made on the day of surgery by the anaesthetist after evaluating the patient.

ASA PS Classification	Definition	Examples, including, but not limited to:
ASA I	A normal healthy patient	Healthy, non-smoking, no or minimal alcohol use
ASA II	A patient with mild systemic disease	Mild diseases only without substantive functional limitations. Examples include (but not limited to): current smoker, social alcohol drinker, pregnancy, obesity (30<BMI<40), well-controlled DM/HTN, mild lung disease
ASA III	A patient with severe systemic disease	Substantive functional limitations; One or more moderate to severe diseases. Examples include (but not limited to): poorly controlled DM or HTN, COPD, morbid obesity (BMI ≥40), active hepatitis, alcohol dependence or abuse, implanted pacemaker, moderate reduction of ejection fraction, ESRD undergoing regularly scheduled dialysis, premature infant PCA < 60 weeks, history (>3 months) of MI, CVA, TIA, or CAD/stents.
ASA IV	A patient with severe systemic disease that is a constant threat to life	Examples include (but not limited to): recent (<3 months) MI, CVA, TIA, or CAD/stents, ongoing cardiac ischemia or severe valve dysfunction, severe reduction of ejection fraction, sepsis, DIC, ARD or ESRD not undergoing regularly scheduled dialysis
ASA V	A moribund patient who is not expected to survive without the operation	Examples include (but not limited to): ruptured abdominal/thoracic aneurysm, massive trauma, intracranial bleed with mass effect, ischemic bowel in the face of significant cardiac pathology or multiple organ/system dysfunction
ASA VI	A declared brain-dead patient whose organs are being removed for donor purposes	
*The addition of "E" denotes Emergency surgery: (An emergency is defined as existing when delay in treatment of the patient would lead to a significant increase in the threat to life or body part)		

Rockwood CFS (Clinical Frailty Scale)
Frailty is a state of increased vulnerability to poor resolution of homoeostasis after a stressor event, which increases the risk of adverse outcomes. It can be assessed quickly and simply using the Rockwood Clinical Frailty Scale (see table below). Frailty identification should take no more the one minute; the more you use the scale, the quicker it will become.

Clinical Frailty Scale*

1 **Very Fit** – People who are robust, active, energetic and motivated. These people commonly exercise regularly. They are among the fittest for their age.

2 **Well** – People who have **no active disease symptoms** but are less fit than category 1. Often, they exercise or are very **active occasionally**, e.g. seasonally.

3 **Managing Well** – People whose **medical problems are well controlled,** but are **not regularly active** beyond routine walking.

4 **Vulnerable** – While **not dependent** on others for daily help, often **symptoms limit activities.** A common complaint is being "slowed up", and/or being tired during the day.

5 **Mildly Frail** – These people often have **more evident slowing,** and need help in **high order IADLs** (finances, transportation, heavy housework, medications). Typically, mild frailty progressively impairs shopping and walking outside alone, meal preparation and housework.

6 **Moderately Frail** – People need help with **all outside activities** and with **keeping house**. Inside, they often have problems with stairs and need **help with bathing** and might need minimal assistance (cuing, standby) with dressing.

7 **Severely Frail** – **Completely dependent for personal care**, from whatever cause (physical or cognitive). Even so, they seem stable and not at high risk of dying (within ~ 6 months).

8 **Very Severely Frail** – Completely dependent, approaching the end of life. Typically, they could not recover even from a minor illness.

9. **Terminally Ill** - Approaching the end of life. This category applies to people with **a life expectancy <6 months**, who are **not otherwise evidently frail.**

Scoring frailty in people with dementia

The degree of frailty corresponds to the degree of dementia. Common **symptoms in mild dementia** include forgetting the details of a recent event, though still remembering the event itself, repeating the same question/story and social withdrawal.

In **moderate dementia**, recent memory is very impaired, even though they seemingly can remember their past life events well. They can do personal care with prompting.

In **severe dementia**, they cannot do personal care without help.

* 1. Canadian Study on Health & Aging, Revised 2008.
2. K. Rockwood et al. A global clinical measure of fitness and frailty in elderly people. CMAJ 2005;173:489-495.

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Online Calculators

Perioperative risk prediction is a rapidly expanding area of research and there are an increasing number of validated online tools and apps available for clinical use. Common examples include:

- Surgical Outcome Risk Tool (SORT)
- P-POSSUM (and specialty specific derivatives)
- American College of Surgeons NSQIP Surgical Risk Calculator

Assessment of Functional Capacity

Patients' exercise tolerance or "functional capacity" is a key determinant of perioperative risk.

The ability to climb stairs (assessed either subjectively from the patient history or objectively through supervised stair climbing +/- pulse oximetry) has long been used to quantify this.

The quantitative and objective assessment of functional capacity is a rapidly expanding field of research. Validated methods of clinical assessment include:

- Duke Activity Status Index (DASI)
- Six minute walk test (6MWT)
- Incremental Shuttle Walk Test
- Cardiopulmonary exercise testing (CPET)

NICE Surgery Grades

The magnitude of the planned surgery (and hence the severity of the physiological insult) is a key determinant of perioperative risk.

NICE surgery grading provides a framework with which to classify this:

Minor	Intermediate	Major or complex
• excising skin lesion • draining breast abscess	• primary repair of inguinal hernia • excising varicose veins in the leg • tonsillectomy or adenotonsillectomy • knee arthroscopy	• total abdominal hysterectomy • endoscopic resection of prostate • lumbar discectomy • thyroidectomy • total joint replacement • lung operations • colonic resection • radical neck dissection